EPANET-JDA (EPANET Junction Demand Allocator)

User Guide & Technical Reference Manual — Version 0.1.0

1. Overview and Capabilities

EPANET-JDA is a Windows-based utility which calculates junction base demands of an EPANET 2.2 hydraulic network model. It supports two standard demand allocation methodologies: **Unit-Length** and **Equal-Distribution** (refer later sections for detail).

The primary purpose of demand allocation is to distribute a network's total demand among its junctions. This is necessary when demands of individual junctions of a network are not known or metered junction demands are not used or the distribution losses need to be added in proportion. When the network is divided into zones (or selection sets), each with its own total demand, the same principle applies at the zone level: each zone's demand is distributed among the junctions belonging to that zone.

Several methods may be used for this allocation. The **Unit-Length** method is one of the most commonly used because it distributes total demands in proportion to the length of pipe connected to each junction. Whereas, **Equal-Distribution** is simple, which distributes total demands equally to each junction.

The program reads a user-supplied input (.inp) file of EPANET 2.2, computes junction demands according to user-defined zones and weighting parameters, and writes a separate, verified input (.inp) file. Existing junction base demands in the **[JUNCTIONS]** section are replaced with the calculated values. Any active records in the **[DEMANDS]** section are removed to eliminate double-counting. Associated demand pattern assignments, operational controls, curves, and link parameters remain untouched. The source file is never overwritten or altered.

The application does not execute EPANET hydraulic simulations and does not require the EPANET simulation engine installed on the host machine.

2. System Requirements and Installation

Operating System: Windows 10 or Windows 11. **Runtime:** Standalone desktop executable (no external installer or runtime setup wizard required). **Package Archive (epanetjda.zip):** Contains epanetjda.exe, epanetjdahelp.pdf, and one EPANET 2.2 sample input (.inp) file.

Extract the entire archive to any local directory.

3. Typical Workflow

3.1 **Export Network File:** In EPANET 2.2, open or verify your network. Navigate to **File** > **Export** > **Network…** and export a clean input (.inp) file.

3.2 **Determine Demand Boundaries:** For **Unit-Length**, compile total zone demands, zone junction lists, and pipe IDs where $K \neq 1.0$. For **Equal-Distribution**, compile total zone demands, zone junction lists, and junction IDs where $N \neq 1.0$.

3.3 **Launch the Program:** Open epanetjda.exe. The initial workspace appears as shown in **Figure 1**.

3.4 **Load .inp file:** Click **Browse…** to select your .inp file. Check the status panel to verify junction counts, link counts, and network flow units.

3.5 **Select Method:** Choose **Unit-Length** or **Equal-Distribution** from the method drop-down list.

3.6 **Set Demand & Units:** Enter the total network demand and confirm that the selected flow unit strictly matches the unit configured in the source file's **[OPTIONS]** section.

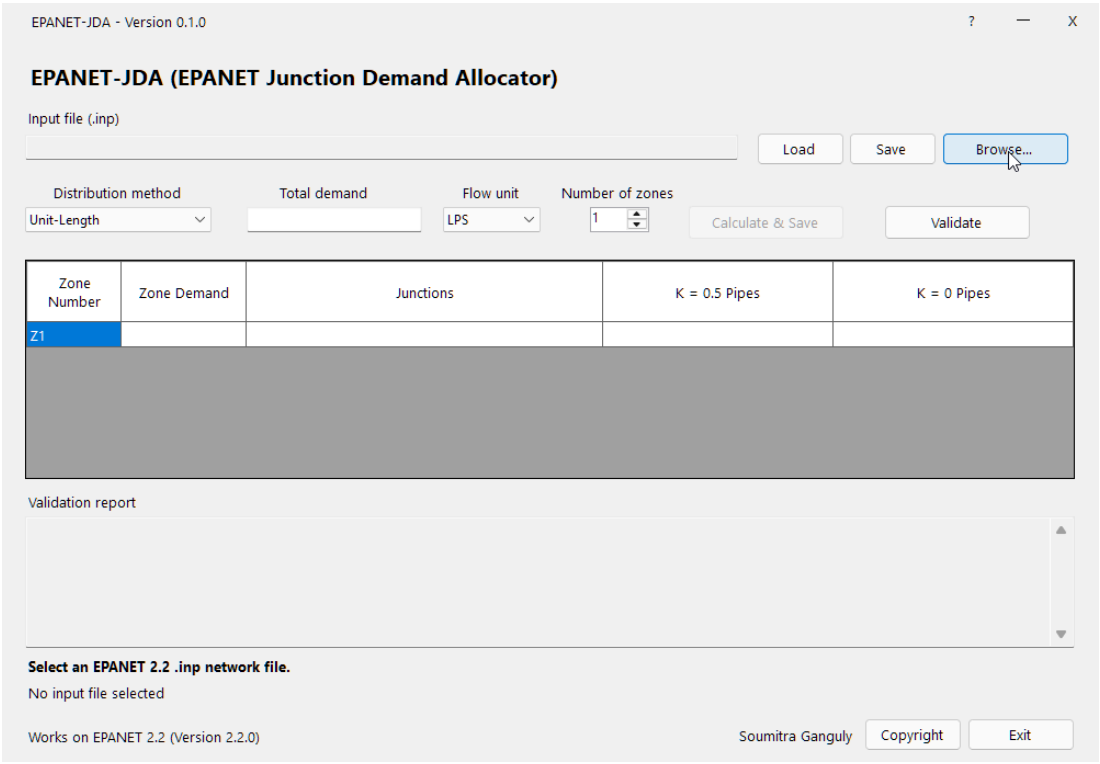


Figure 1: Main window interface before selecting a network file

Ensure that, when the Browse… button is highlighted (receives focus), a help message is displayed.

3.7 **Define Zones:** Enter the number of zones; matching rows will automatically populate in the Zone Detail grid.

3.8 **Populate Zone Data:** Provide unique zone names, target zone demands, and assigned Junction IDs (using comma-separated values and/or numeric ranges like J1-J5, J7, J9).

3.9 **Assign Modifiers:**

3.9.1 Under **Unit-Length**, enter pipe IDs requiring K = 0.5 or K = 0.0.

3.9.2 Under **Equal-Distribution**, enter junction IDs requiring N = 0.5 or N = 0.0.

3.10 **Validate Inputs:** Click **Validate**. Correct any reported errors in the validation pane. Validation verifies network topology and demand balances without modifying files.

3.11 **Calculate & Save:** Click **Calculate & Save**. The program computes allocations, generates a new .inp file, and also saves an analysis report (.txt) in the same folder as the source file. The output filename appears in the window; make a note of it.

3.12 **File Versioning:** Generated files are automatically saved with the original name plus name of the method used adding sequential tags (e.g., Sample1_UnitLength_1.inp or Sample1_EqualDistribution_1.inp). Existing input files are never overwritten.

3.13 **Session State (Optional):** Click **Save** to export current workspace settings to a .dgsession.json session file. Click **Load** to restore an earlier session.

3.14 **Hydraulic Simulation:** Open the resulting .txt to inspect base demands. Finally, import the .inp file into EPANET 2.2, verify base demands, and run network simulations.

The application does not run a hydraulic simulation.

The complete build and run instructions are also available in the project README file.

4. Input Rules and Validation Constraints

4.1 Network Total and Flow Units

Network total demand must be a finite, positive number. Supported EPANET flow units: CFS, GPM, MGD, IMGD, AFD, LPS, LPM, MLD, CMH, and CMD. The selected unit must match the unit declared in the network's **[OPTIONS]** section. The sum of all zone demands must reconcile with the network total demand within a tolerance of 0.1%. Discrepancies exceeding 0.1% are flagged as fatal validation errors; minor rounding variances trigger informational warnings.

4.2 Demand Pattern and Section Handling

[JUNCTIONS] Section: The base demand column (Column 3) of the .inp file is updated with calculated values. If an existing junction entry references a Demand Pattern ID (Column 4), that pattern association remains preserved. **[DEMANDS]** Section: Any secondary or extended demand records are removed to prevent duplicating consumer baseline drafts. Original .inp: The original .inp file remains completely untouched.

4.3 Zone and Element Constraints

Zone identifiers must begin with an alphabetic character, may contain alphanumeric characters, hyphens, or underscores, and must not exceed 31 characters. Every junction in the network must be assigned. Unassigned junctions or duplicate junction assignments between zones will trigger validation failures. Reservoirs and tanks shares no demand and cannot be assigned to zones. Non-pipe link elements (pumps, PRVs, PSVs, PBVs, FCVs, TCVs) possess no physical pipeline length and are omitted from Unit-Length weighting calculations.

5. Method-specific modifier columns

The last two grid columns have different meanings depending on the selected method.

Method	Column 1 modifier	Column 2 modifier	Default (Unlisted)
Unit-Length	Pipe IDs with K = 0.5	Pipe IDs with K = 0	K = 1.0 (Full weight)
Equal-Distribution	Junction IDs with N = 0.5 (50%)	Junction IDs with N = 0.0 (0%)	N = 1.0 (100% weight)

For **Unit-Length**, unlisted pipes use K = 1. A pipe modifier must be entered under a zone that contains at least one endpoint of that pipe. A pipe can have only one K assignment. If a pipe belongs to two zones, its ID can be entered in either zone, but not both.

A valve or pump is a link without a pipe length and is not used by Unit-Length length weighting. For example, the valve V1 in the sample network connects J2 and J6; it has no length contribution. J6 remains a normal junction assigned to Z2.

Every positive-demand Unit-Length zone must have positive K-weighted pipe length contributing to its junctions. A pipe with K = 0 contributes no weight to either endpoint.

For **Equal-Distribution**, unlisted junctions use 100% weight. A positive-demand zone must retain at least one junction with positive weight; it cannot assign 0% weight to every junction.

6. Calculation methods

6.1 Unit-Length

Unit-Length allocates each zone's entered demand in proportion to the K-weighted pipe length assigned to its junctions.

The underlying assumption is that population—and therefore water consumption—is generally located on both sides of a pipeline. Since pipelines often run along roads, it is reasonable to assume that demand is spread uniformly along the pipe length. Under this assumption, the total demand is distributed uniformly over the entire pipe network, giving a uniform demand per metre of pipe.

Each junction is then considered to "own" half of the length of every pipe connected to it. Thus, a junction connected to three pipes receives half of the effective length of each of those three pipes. Its share of the zone demand is proportional to the total effective pipe length assigned to it.

In practice, however, this assumption needs refinement. Not every pipe has population on both sides of it. Some pipes mainly convey water from one part of the network to another and have little or no associated consumption. The K factor addresses this by allowing the user to reduce, or completely remove, a pipe's contribution to demand allocation. A pipe with K=1 contributes its full length; a pipe with K=0.5 contributes half its length; and a pipe with K=0 contributes no length, even though it remains part of the hydraulic network.

Zone boundaries are represented by junctions, not by pipes. Therefore, when a pipe connects junctions in two different zones, its effective length is divided equally between the two zones: half is assigned to the junction in one zone and half to the junction in the other. Each zone then normalizes the demands of its own junctions against its own total effective length, ensuring that the full entered demand of each zone is preserved.

The central idea of the Unit-Length method is therefore to distribute demand in a more realistic and physically meaningful way: junctions receive demand in proportion to the weighted pipe length they share, while the user can adjust individual pipes to reflect the actual presence or absence of population and consumption along them.

The K factor changes a pipe's *weight in the allocation*; it does not directly set a junction's demand or reduce the zone's entered demand.

Meaning of K and effective pipe length

Each pipe has a physical length L. Its K factor is a dimensionless multiplier:

$$L_{\text{eff}} = KL$$

- **K = 1.0** (the default): the full pipe length contributes to the weighting.
- **K = 0.5**: half the pipe length contributes.
- **K = 0**: none of the pipe length contributes. The pipe still exists in the network, but it gives no Unit-Length weight to either junction.

Pipe with both junctions in the same zone

If a pipe p connects junctions a and b, and both belong to the same zone Z, its effective length is divided equally between them:

$$w_{a,p} = w_{b,p} = \frac{K_p L_p}{2}$$

Pipe K	Weight added at junction a	Weight added at junction b	Total added to zone Z
1.0	L/2	L/2	L
0.5	L/4	L/4	L/2
0	0	0	0

These are weights, not demands. Each junction's final weight includes contributions from *all* its incident pipes. If W_z is the sum of the weights at every junction in zone Z, then:

$$D_a = D_z \frac{w_a}{W_z}, \qquad D_b = D_z \frac{w_b}{W_z}$$

Pipe with junctions in two different zones

If the junction a is in zone Z1 and junction b is in zone Z2 connected by pipe p, the pipe is still split equally. Each zone receives only the half assigned to its own junction:

$$w_{a,p} = \frac{K_p L_p}{2}, \qquad w_{b,p} = \frac{K_p L_p}{2}$$

Pipe K	Weight added to zone Z1 at a	Weight added to zone Z2 at b
1.0	L/2	L/2
0.5	L/4	L/4
0	0	0

The two junctions are normalized against their *own zone totals*, which may differ:

$$D_a = D_{Z1} \frac{w_a}{W_{Z1}}, \qquad D_b = D_{Z2} \frac{w_b}{W_{Z2}}$$

Pipe connected to a reservoir or tank

Reservoirs and tanks are not junctions and do not receive demand or allocation weight. If endpoint a is a reservoir or tank and endpoint b is a junction, the junction receives the pipe's *full* effective length:

$$w_{a,p} = 0, \qquad w_{b,p} = K_p L_p$$

Pipe K	Weight added at reservoir/tank a	Weight added at junction b
1.0	0	L
0.5	0	L/2
0	0	0

The junction's demand is then $D_b = D_z \times w_b / W_z$. If this is the only pipe contributing weight in the zone and K is positive, b receives the entire zone demand.

Effective length total for a zone

The total effective pipe length for zone Z is the sum of the weights at its junctions:

$$w_j = \sum_{p \text{ incident to } j} w_{j,p}, \qquad W_z = \sum_{j \in J_z} w_j, \qquad D_j = D_z \frac{w_j}{W_z}$$

This zone total is **not generally the same as the physical total length of pipes one might associate with that zone**. K can reduce a pipe's contribution, and a cross-zone pipe is split rather than counted in full for both zones. For example, consider:

A positive-demand zone must have $W_z > 0$.

The application adjusts floating-point rounding at one junction so the calculated demands sum to the entered zone demand.

6.2 Equal-Distribution

The **Equal-Distribution** method is the simplest approach: it distributes demand equally among the eligible junctions, rather than estimating demand from pipe length, population, or any other network property.

The allocation weight expresses the junction's share in the distribution. A junction with weight 1.0 receives a full equal share; a junction with weight 0.5 receives half of a full share; and a junction with weight 0 receives no demand.

For each junction j, let a_j be its allocation weight. For zone Z:

$$A_z = \sum_{j \in J_z} a_j, \qquad D_j = D_z \frac{a_j}{A_z}$$

The calculated junction demands within each zone always sum to the zone's entered demand. A zone with zero entered demand may have zero allocation weights throughout, since no demand needs to be distributed.

7. Worked example: sample network

The package includes Sample1.inp. The network contains 10 junctions (J1 through J10), one reservoir (R1), one valve (V1 connecting J2 to J6), 10 pipes, and the flow unit LPS with total network demand defined at 50 LPS.

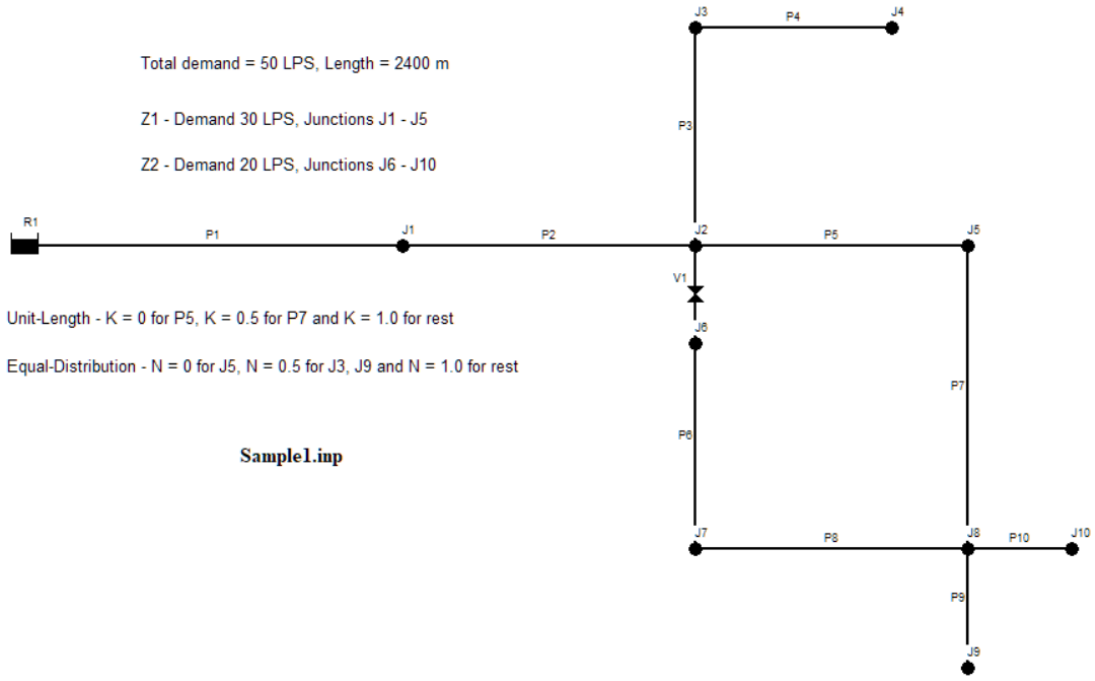


Figure 2: Sample1 network file

The valve connects J2 to J6. It has no pipe length and does not contribute to Unit-Length weighting. The valve is a separate link and must not be entered as a pipe modifier.

The system is evaluated under a two-zone configuration:

Unit-Length:

Zone	Demand	Junctions	Pipe K Inputs
Z1	30 LPS	J1-J5	K = 0 for P5
Z2	20 LPS	J6-J10	K = 0.5 for P7

Equal-Distribution:

Zone	Demand	Junctions	Junction N Inputs
Z1	30 LPS	J1-J5	N = 0.5 for J3 and N = 0 for J5
Z2	20 LPS	J6-J10	N = 0.5 for J9

The two zones cover all 10 junctions exactly once.

7.1 Unit-Length example

Set demand allocation for Z1 to 30 LPS and for Z2 to 20 LPS. For Z1, set P5 to K = 0. All other pipes use K = 1 by default. For Z2, set P7 to K = 0.5. All other pipes use K = 1 by default. A pipe connected to a reservoir or tank contributes its full effective length to its junction endpoint. A pipe

between two junctions contributes half of its effective length to each endpoint.

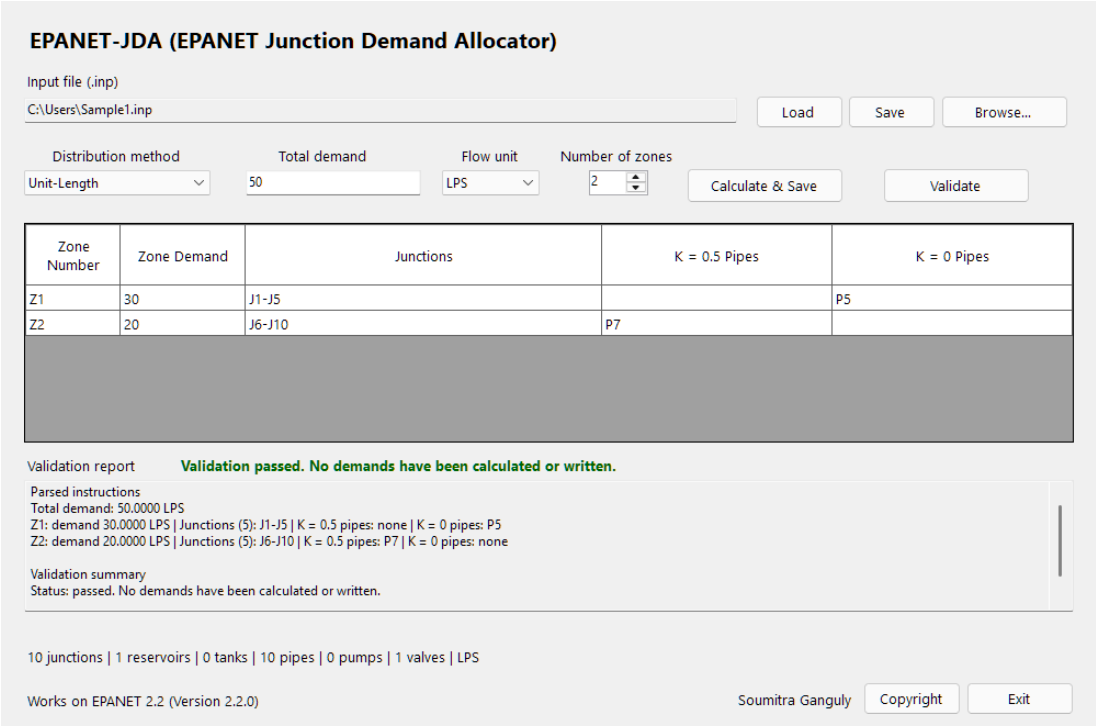


Figure 3: Main window after entering Unit-Length data

Zone	Junction	Weight	Allocation Ratio	Demand (LPS)	Notes
Z1	J1	550.0	550.0/1177.5	14.0127	P1 (400 m), P2 (150 m)
Z1	J2	265.0	265.0/1177.5	6.7516	P2 (150 m), P3 (115 m), V1 (0 m)
Z1	J3	200.0	200.0/1177.5	5.0955	P3 (115 m), P4 (85 m)
Z1	J4	85.0	85.0/1177.5	2.1656	P4 (85 m), P5 (0 m, K=0)
Z1	J5	77.5	77.5/1177.5	1.9745	P5 (0 m), P7 (77.5 m, K=0.5)
Z1 Total	–	1177.5	–	30.0	Zonal demand matched
Z2	J6	155.0	155.0/817.5	3.7920	V1 (0 m), P6 (155 m)
Z2	J7	280.0	280.0/817.5	6.8502	P6 (155 m), P8 (125 m)
Z2	J8	292.5	292.5/817.5	7.1560	P7 (77.5 m), P8 (125 m), P9 (50 m), P10 (40 m)
Z2	J9	50.0	50.0/817.5	1.2232	P9 (50 m, dead-end branch)
Z2	J10	40.0	40.0/817.5	0.9786	P10 (40 m, dead-end branch)
Z2 Total	–	817.5	–	20.0	Zonal demand matched
Network	–	1995.0	–	50.0	Total demand matched

The total Unit-Length weight is 1,177.5 m for Z1 and 817.5 m for Z2. The values are rounded to four decimal places, while the application calculates with full precision.

For example, P1 contributes 400 to J1 because it connects the reservoir R1 to J1. P2 contributes 150 to both J1 and J2. P7 contributes 77.5 to J5 and J8 because its effective length is 155 × 0.5. P5 contributes nothing because its K factor is 0.

7.2 Equal-Distribution example

Set demand allocation for Z1 to 30 LPS and for Z2 to 20 LPS.
For Z1, set J3 to N = 0.5 (50%) and J5 to N = 0 (0%). All other junctions use N = 1 by default. The resulting Z1 total weight is 3.5. For Z2, set J9 to N = 0.5 (50%). All other junctions use N = 1 by default. The resulting Z2 total weight is 4.5.

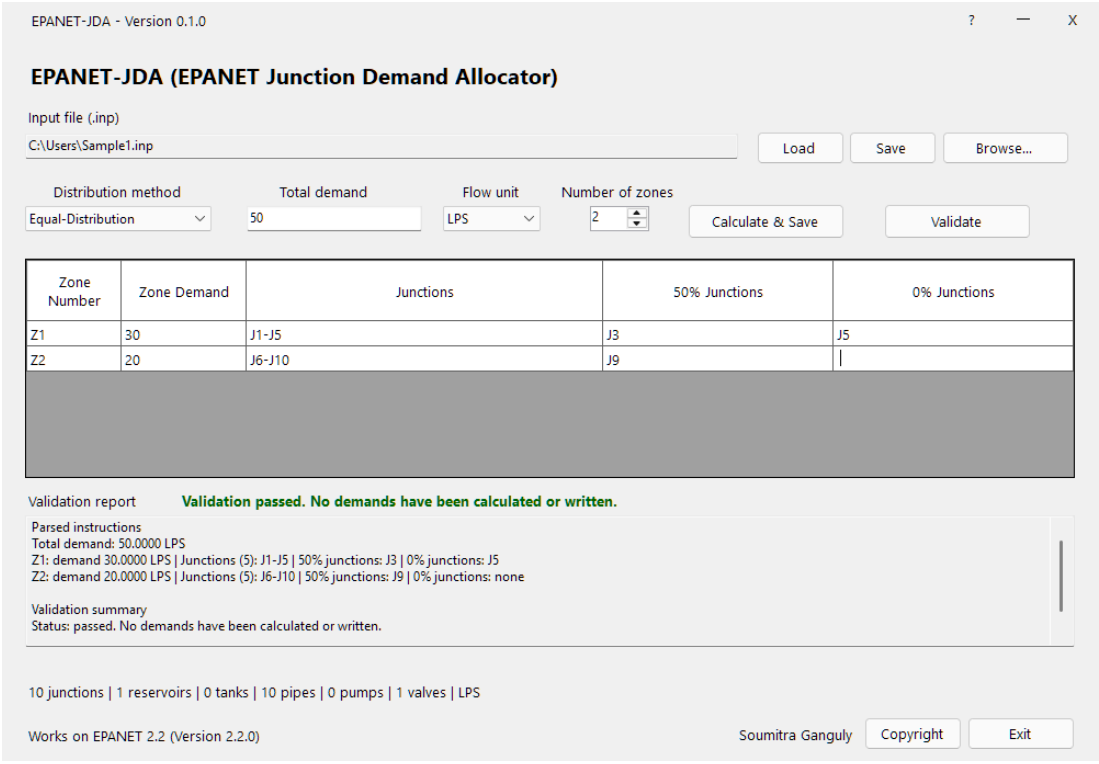


Figure 4: Main window after entering Equal-Distribution data

Zone	Junction	Weight	Allocation Ratio	Allocated Demand (LPS)	Notes
Z1	J1	1.0	1.0 / 3.5	8.5714	Standard junction
Z1	J2	1.0	1.0 / 3.5	8.5714	Standard junction
Z1	J3	0.5	0.5 / 3.5	4.2857	50% modifier applied
Z1	J4	1.0	1.0 / 3.5	8.5714	Standard junction
Z1	J5	0.0	0.0 / 3.5	0.0000	0% modifier applied
Z1 Total	–	3.5	–	30.0	Zonal demand matched
Z2	J6	1.0	1.0 / 4.5	4.4444	Standard junction
Z2	J7	1.0	1.0 / 4.5	4.4444	Standard junction
Z2	J8	1.0	1.0 / 4.5	4.4444	Standard junction
Z2	J9	0.5	0.5 / 4.5	2.2222	50% modifier applied
Z2	J10	1.0	1.0 / 4.5	4.4444	Standard junction
Z2 Total	–	4.5	–	20.0	Zonal demand matched
Network	–	8.0	–	50.0	Network demand matched

8. Reviewing generated files

The detailed report file (.txt) lists the network summary, entered instructions, calculated junction demands and weights, warnings, and entered-versus-allocated totals. Check that the zone and network totals match the intended values.

The generated input (.inp) file contains the new junction base demands. Active **[DEMANDS]** records from the source are removed to prevent double counting. The source input file is not modified.

If generation fails, read the validation or error text in the application, correct the source or entered values, validate again, and generate a new copy. Existing generated files and reports are preserved; a new sequence number is used rather than overwriting them.

9. License

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